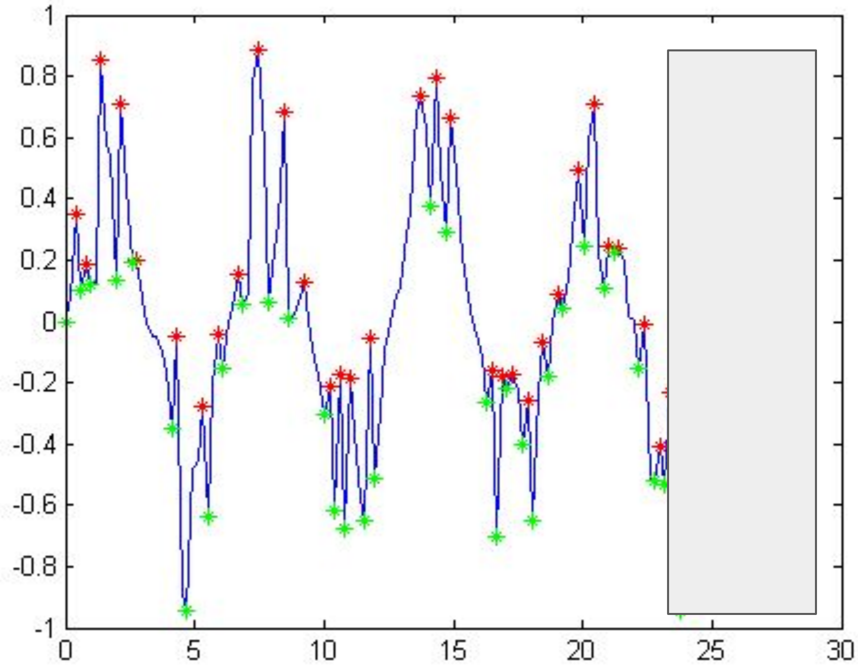
Physics Stumping Strategies and Resources

How the Models Work



Stumping the Models

There are two things to keep in mind when trying to find areas in which the models perform poorly:

1. An AI model's success is a **strong function of the volume of similar training data it has seen**.
    a. There are many areas that, while well-established in research literature, do not make it into textbooks. **These are good areas to probe the model in.**

2. AI models tend to be **very bad at extrapolating to novel contexts**.
    a. This can include applying a framework/approximation typically used in one scenario in an entirely distinct scenario (in which it still applies).
    b. It can also include breaking standard approximations/assumptions.
        i. In this case, you can design the prompt so that the parameter values/context preclude an approximation without explicitly telling the model not to use it. **If it applies the approximation anyways, this is a valid failure (but it can't be borderline-must clearly not apply).**

[Physics Stumping Strategies and Resources Link]

Techniques that don’t work

Hyper-complex calculations

Textbook like problems

Using a result/equation from a single paper

Trickery-things that don’t apply

Contradictions

Good Stumping Strategies

Niche Areas

Frameworks applied out of context

Tractable but Novel Derivations

Implicitly Broken Approximation

Implicit Data/Complicated physical reasoning

Experimental set-ups

Add boundary conditions: Under which case, use certain values/methods

Good Stumping Strategies

Add more reasoning layers: stitch more concepts and approaches into the task

Use stochastic rather than deterministic: Gaussian noise, uniform probability

Combine topics that aren’t often combined

Start standard, make a big change that causes something fundamentally different

Deductive reasoning	Drawing specific conclusions from general laws or principles.	E.g., "Given this mechanism, what must follow?"
Inductive reasoning	Generalizing from patterns or experimental observations.	E.g., "Given this data, what can we infer about X?"
Temporal reasoning	Predicting events or states based on order in time.	E.g., "If A happens before B, and B affects C, what's the downstream effect?"
Spatial reasoning	Understanding structures, orientation, or symmetry.	E.g., "Which product isomers exhibit C2v symmetry?"
Causal reasoning	Identifying cause-and-effect relationships.	E.g., "What change in gene expression could cause this phenotype?"
Comparative analysis	Judging between alternatives or evaluating differences across conditions.	E.g., "Which treatment has higher efficacy and why?"
Abstract reasoning	Working with non-concrete or theoretical ideas.	E.g., "What's the entropy change in a hypothetical closed system with constraint X?"
Pattern recognition	Spotting and interpreting regularities in data or sequences.	E.g., "Which degenerate codons could result in histidine insertion?"
Statistical reasoning	Using data, probabilities, and distributions to reach conclusions.	E.g., "What's the statistical likelihood that the library diversity meets the threshold?"
Abductive reasoning	Inferring the most likely explanation from incomplete evidence.	E.g., "What's the explanation for this anomaly in the assay results?"
Hypothetical reasoning	Predicting outcomes under hypothetical or counterfactual scenarios.	E.g., "If we changed the methylation site, how will binding affinity shift?"

How one specialist builds tasks

- **Identify my areas of expertise.** There are many areas I'm competent in, and many questions I could answer if given enough time. However, there are really only a few topics in which I'd consider myself truly an expert – i.e., in which I can fluently communicate at the level of a researcher or specialist. For me, these are stratified environmental flows, nanofluidics, nonlinear response theory, and chiral active fluids. Note that these topics are few, niche, and specific.
- **Identify what I think the model would be bad at.** […]
- **What is a problem that interest me, but I don't know the answer to?** […]
  - Here, I might break common assumptions just to see what will happen.
  - I might try and apply something I know to a completely novel context that I've thought up.
  - I might smash two apparently distinct/unrelated topics together.

How a different specialist builds taks

1. Pick an area of interest
2. Start with a general open-ended question (possibly at undergrad level)
3. Analyze responses and ask another more focused question at an area that might show signs of weakness (if there are any)
4. Repeat process, probing for weakness. Once that is found, ask the detailed question that follows the guidelines

Niche Research Areas

- **Explanation:** There are topics that are well-established in academic research but are too niche or specific in their application to merit inclusion in textbooks/curricula. These results often exist only in primary research and review papers.
- **Example:**
  - **Topic:**Anthony studied a field called two-layer hydraulics, which governs the density-driven flow of two (approximately) immiscible fluids.
    - This field is **well-established**. Original results go back 80+ years, with the primary theoretical foundation being laid in the 1980s.
    - It is **uncontroversial**. No expert familiar with its theoretical foundations finds them objectionable.
    - It has **limited applications**. Consequently, there are few (or possibly no) textbooks that treat the subject.

Frameworks Applied Out-of-Context

- **Explanation:** The model is very good at applying the correct theoretical framework in the context it is usually applied. However, you can sometimes design prompts where the same framework applies but the context is different. A good example is the two-layer stratified flow example above (**Niche Research Areas**). Another good example is the turbulence scaling problem presented in the **Tractable But Novel Derivations** section below.

Tractable But Novel Derivations

- **Explanation:** Research papers are often a good source of derivations that start from uncontroversial initial points and proceed through verifiable logical and mathematical steps, but are advanced or niche enough that the models are unlikely to have seen them. These derivations cannot be used directly, but they are good starting points for distinct but similar calculations.
- **Example:**
    - **Topic:** In Anthony's research, he derived a novel system of equations to describe the transport of ions in nanochannels/nanopores in a specific limiting regime.
        - The derivation proceeds from **common, universally accepted starting assumptions**.
        - Though it may take a bit of effort, **any expert in the field could derive the equations, and all could verify their correctness if presented with the derivation**.
        - Nonetheless, we know that they are a relatively recent addition to the literature and are **unlikely to be significantly represented in the model's training data**.
        - Of course, you don't have to rely on research you've done. This can be inspired by any research, as long as **the framework is uncontroversial, the starting point is clearly given**, **the derivation is logically and mathematically contained/verifiable**, and **the requested calculation/derivation is distinct** from what is found in the paper.

Implicitly Broken Approximations

- **Explanation:** Sometimes, classes of problems are almost universally solved by making simplifying approximations. Some examples are the small angle approximation in physics, the small deflection approximation in beam analysis, and the small density-difference approximation in stratified flows. You can design prompts with **parameter values that unambiguously exclude these approximations**. If the model makes them anyway, this is a valid failure.
    - **Note:** The invalidity of the assumption must be truly unambiguous. This does not mean small quantitative differences between an approximate and an exact solution. It means **truly divergent results that are due to the physical absurdity of applying the approximation**.

Implicit Data/Complicated Physical/Engineering Reasoning

- **Explanation:** Failures can sometimes be induced by paring down the information given in the prompt to the bare minimum. Rather than giving every parameter value as an explicit numerical value, you can force the model to infer withheld information using physical/engineering reasoning. You can also rely on cancellations during the solution procedure that render nominally important information irrelevant.
- **Example:**

Don’t Use Specific, Empirical Findings from a Single Study!

- You can use papers for inspiration.
    - This should only be for calculations.
    - An expert in your field should be able to verify your solution without having read the paper.
    - The models won’t reference the paper, even if you provide it as a source. Therefore, a hypothetical expert shouldn’t need to either.
- These are the standards for a calculation inspired by a research paper:
    - The same calculation/derivation cannot be found in the paper.
        - You can derive a similar quantity under substantially different assumptions.
        - You can push through a basic theoretical result to obtain a specific numerical result for a given system.
    - The starting point of the calculation or derivation must be uncontroversial.
        - If experts in your field would disagree as to whether the starting framework/assumptions are valid/applicable, this is not appropriate. Experts in your field have to agree on the validity of your solution.
    - The starting point must be clearly given.
        - If the model must start from a specific equation or framework, this should be explicitly stated or given in the prompt.
    - Moving from the starting assumptions to the final answer must be deductively verifiable.
        - Every logical step and mathematical manipulation must be such that any expert in your field would agree.